ZEBRAFISH KNOWLEDGE SERIES

The Impact of Zebrafish Research

What a small fish has contributed to medicine, regeneration and environmental science

A summary of the main areas where zebrafish research has made a difference, with a timeline of milestones from the first genetic work at Oregon to the reference genome and drug screening.

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1 Summary

Work on zebrafish has changed how researchers study regeneration, cancer, inherited disease and drug action.

Because the embryo is transparent and the animal reproduces quickly, many questions that are slow or costly in mammals can be answered in days in zebrafish. This paper summarises the main areas where that has made a difference, and lists the milestones that shaped the field.

2 Milestones

- **1970s–80s** George Streisinger and colleagues at the University of Oregon develop the zebrafish as a genetic model. Their clones are among the earliest successful vertebrate clones.
- **1990s** Large-scale forward genetic screens in Tübingen and Boston identify mutants affecting most organ systems.
- **2007** A small-molecule screen in embryos identifies dorsomorphin, an inhibitor of BMP signalling, and a modified fish is reported that detects oestrogen-like compounds in water.
- **2008** The transparent *casper* strain allows tumours and cells to be imaged in adult fish.
- **2011** Transgenic melanoma models carrying BRAF or NRAS mutations are used in screens reported in *Nature*.
- **2012** Spinal cord repair in zebrafish is described in detail.
- **2013** The reference genome is published in *Nature*.
- **2015** A review counts 65 small-molecule screens carried out in zebrafish, at least one of which led to clinical trials.
- **2019** A transparent, immunodeficient strain is reported, enabling the study of transplanted human cells.

Sources: Wikipedia, "Zebrafish", summarising the primary papers; Vanderbilt University Medical Center, 2010.

3 Regeneration

Unlike mammals, zebrafish can regrow tissues that humans cannot. Researchers have documented regeneration of the heart, of the sensory hair cells of the lateral line, of the retina and of the spinal cord. Finding which genes switch on after injury in the fish gives clues about why the same repair does not happen in people, and whether it could be encouraged.

The same approach has been applied to the pancreas, where zebrafish are used to study how insulin-producing beta cells are replaced, which is relevant to diabetes research.

4 Cancer

Cancer models in zebrafish carry the same cancer-causing mutations found in human tumours. Melanoma models using BRAF or NRAS mutations were used in screens published in *Nature* in 2011. The transparent *casper* fish lets researchers see tumour cells spread and settle in a living adult, a process that cannot easily be observed in mice.

5 Drug discovery

Testing a compound in a cell culture shows whether it hits a target. Testing it in a zebrafish embryo also shows whether it reaches the tissue and whether it disrupts development elsewhere in the body. In a Vanderbilt University study, a 2007 embryo screen identified dorsomorphin, which blocks BMP signalling but also affects the VEGF receptor and blood vessel formation. Chemists then made related compounds, and testing them in embryos separated those that were selective for BMP from those that were selective for VEGF. One VEGF inhibitor performed better than a comparison compound that had been dropped from a Phase III cancer trial.

> ### Speed is the point
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> The researchers reported that the embryo assay takes about a day, where the traditional route of testing in cells and then animals takes months. By 2015, 65 small-molecule screens had been run in zebrafish, and at least one had led to clinical trials.

6 Environment and toxicology

Zebrafish embryos absorb chemicals from the water they sit in, so they respond quickly to pollutants. A modified fish that responds to oestrogen-like compounds, first reported in 2007, shows how the animal can serve as a living sensor for water quality as well as a laboratory model.

7 Space biology

Zebrafish are among the few fish that have been flown into space, where they have been used to study how weightlessness affects living organisms.

8 What the impact depends on

Much of this work has used a small number of laboratory strains. Studies of wild zebrafish from Nepal, India and Bangladesh show that these strains carry only part of the variation found in nature (Whiteley et al., 2011). Future discoveries will draw on that natural variation, which is held in wild populations and in the rivers and wetlands where they live.

Sources

Wikipedia: Zebrafish (en.wikipedia.org/wiki/Zebrafish). Used as a summary of primary papers on regeneration, cancer models, strains and screens, including the 2015 review of 65 screens.

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Whiteley A.R. et al. (2011). Population genomics of wild and laboratory zebrafish (*Danio rerio*). *Molecular Ecology* 20(20): 4259–4276.